

ASSIGNMENT 2 – ROOF ANALYSIS

Analysing and improving a multi-layer building component for a roof in the cold climate of Sweden with the simulation tool WUFI PRO 6.8 and the mould growth model VTT

Nahla Alhamada

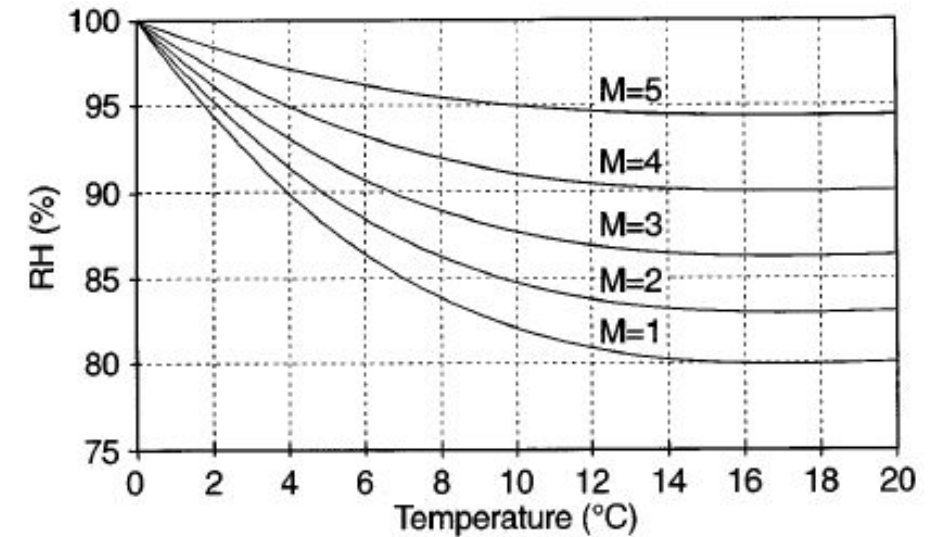
Vinay Dilip

Thomas Obermair

Group 1, AS24, VBFN15 Moisture Safety Design of Buildings
Department of Building and Environmental Technology
Division of Energy and Building Design

INTRODUCTION

- VTT-Mould model
- Critical requirements mould growth:
 - $RH > 80\%$
 - $5^{\circ}\text{C} < T < 50^{\circ}\text{C}$
 - exposure over time
- Mould Index
 - > 1 ... initial stage
 - > 3 ... visible to the eye



Source: Viitannen et.al. (1999)

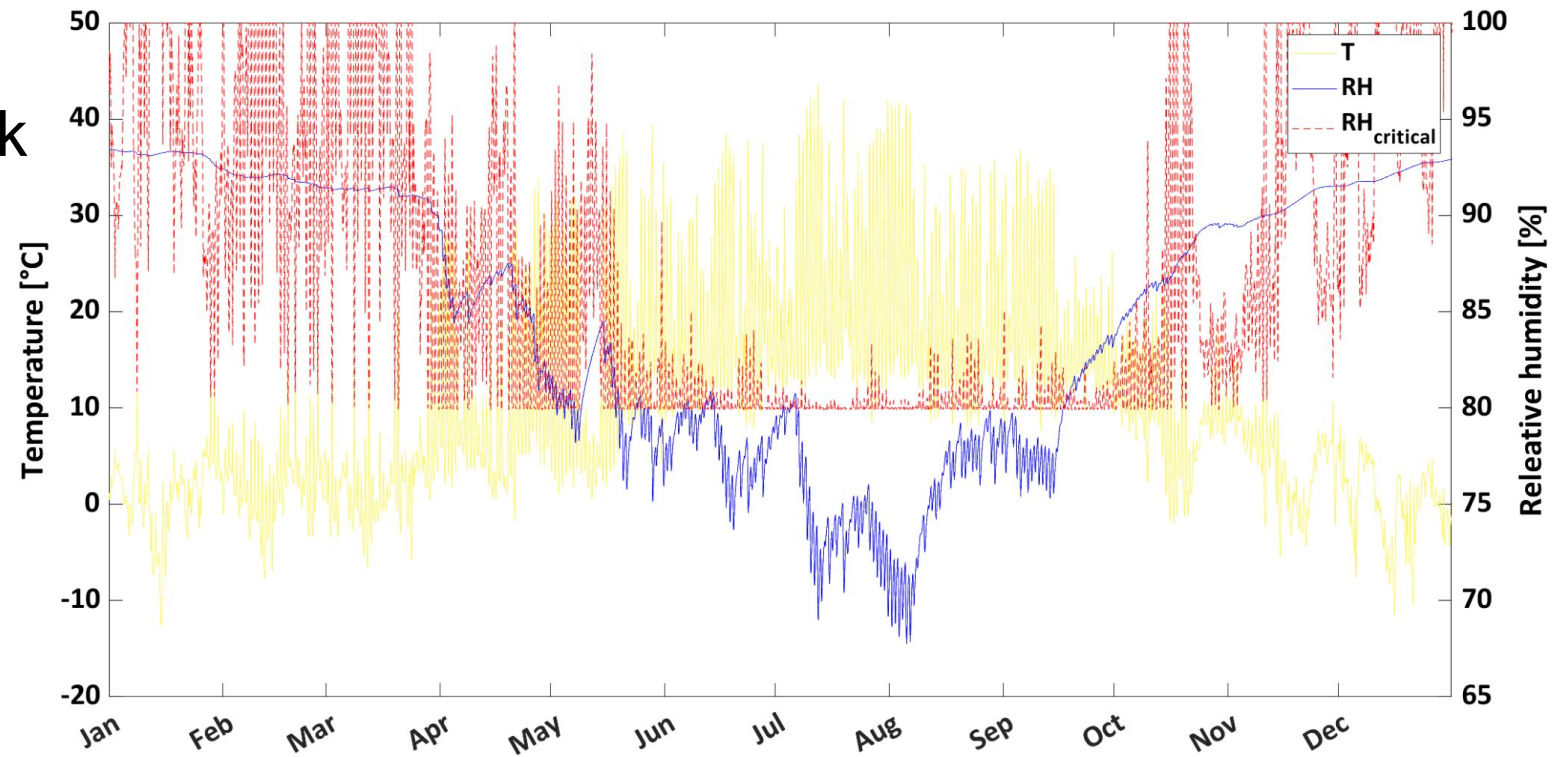
Roof Base Case

Total Thickness 0.406 m
U-Value 0.111 W/m²K

Nr.	s [mm]	Name	Sketch
0		Aluminium Siding ($s_d=50\text{m}$)	
1	22	Spruce radial	
2a	10	Air layer 10 mm	
2b	20	Air layer 20 mm (without additional moisture capacity)	
2c	10	Air layer 10 mm	
3	30	Wood-Fibre Insulation Board	
4	300	Mineral Wool ($\lambda=0.04\text{ W/mK}$; between wooden rafters)	
5	1	Vapour retarder ($s_d=100\text{m}$)	
6	12.5	Gypsum board	

RESULTS BASE CASE

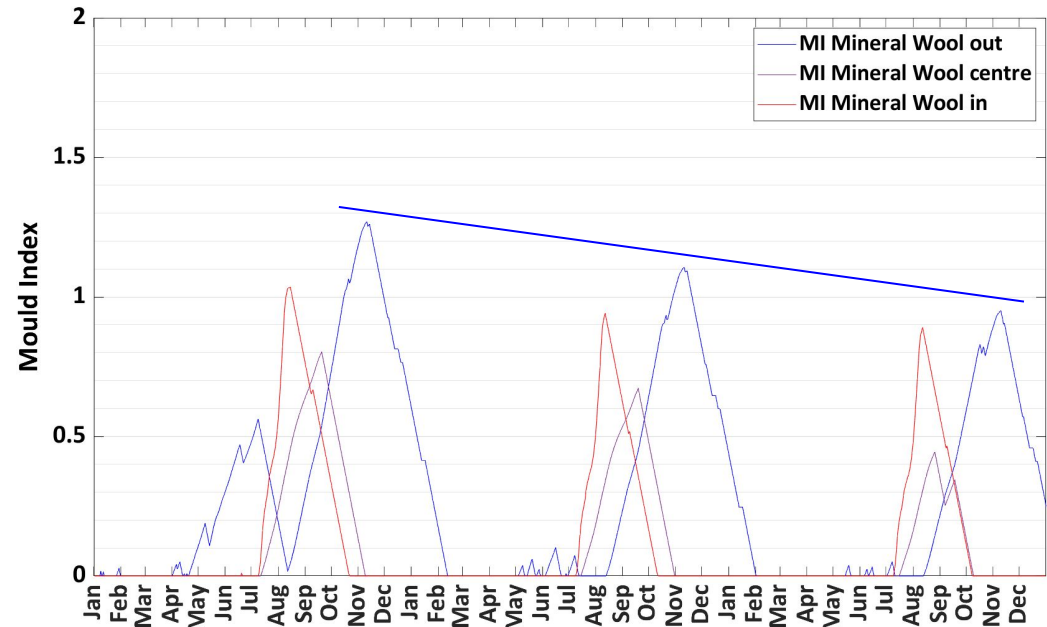
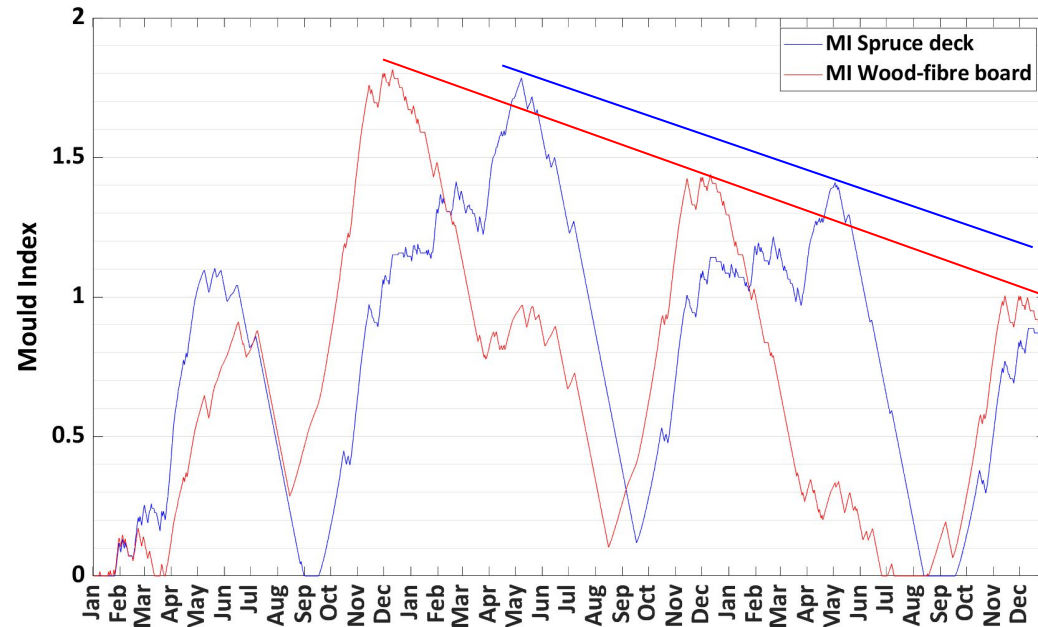
- Sensible layers
 - Layer 1 - Spruce deck



RESULTS BASE CASE

- Sensible layers
 - Layer 1 - Spruce deck
 - Layer 3 - Wood-fibre insulation board
 - Layer 4 - Mineral Wool (worst conditions in outermost point)

RESULTS BASE CASE



Tab. 1 Summary of Mould Growth Assessment for the sensible layers. Base case, three years, incl. initial conditions.

Layer	RH _{max} [%]	RH _{mean} [%]	T _{mean} [°C]	MI _{max}	MI _{mean}	MGR <0 [h-%]
Spruce deck	93.97	85.39	9.95	1.78	0.79	47.21
Wood-Fibre insulation	92.34	84.44	10.55	1.81	0.70	44.20
Mineral Wool (outside)	90.69	81.16	11.48	1.27	0.30	59.73

IMPROVEMENTS

Roof Improved

Total Thickness 0.407 m
U-Value 0.115 W/m²K

Nr.	s [mm]	Name	Sketch
0		Aluminium Siding ($s_d=50m$)	
1	22	Spruce radial	
2a	10	Air layer 10 mm	
2b	20	Air layer 20 mm (without additional moisture capacity)	
2c	10	Air layer 10 mm	
3	1	TRASPIR EVO 160 ($s_d=0,1m$)	
4	35	AiF Rigid Underlay Board WF high density	
5	280	Mineral Wool ($\lambda=0.04$ W/mK; between wooden rafters)	
6	15	AiF OSB 3	
7	1	Vapour retarder ($s_d=100m$)	
8	12.5	Gypsum board	

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8	12.5	Gypsum board	

DOES IT WORK IN THE FUTURE?

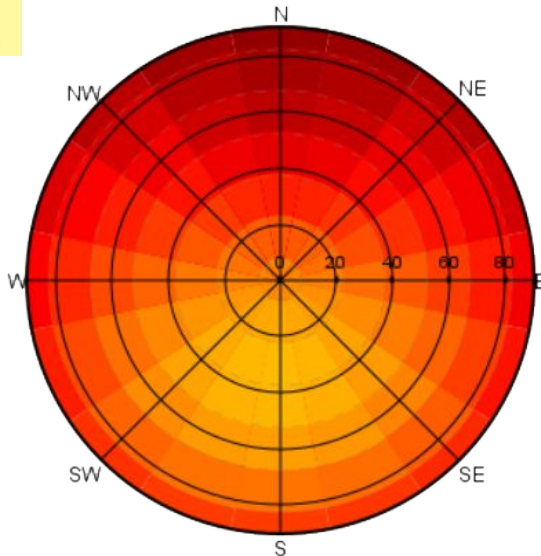
IMPACT OF CLIMATE CHANGE SCENARIO

Base Case

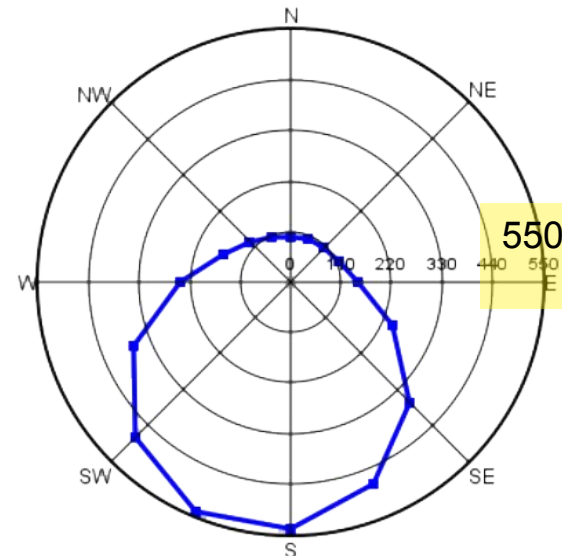
Mean Temperature [°C]:	8.8	Mean Relative Humidity [%]:	74
Max. Temperature [°C]:	27.8	Max. Relative Humidity [%]:	94
Min. Temperature [°C]:	-12.2	Min. Relative Humidity [%]:	19
Mean Wind Speed [m/s]:		3	
Normal Rain Sum [mm/a]:		1073.5	

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Solar Radiation Sum [kWh/m²a]



Driving Rain Sum [mm/a]

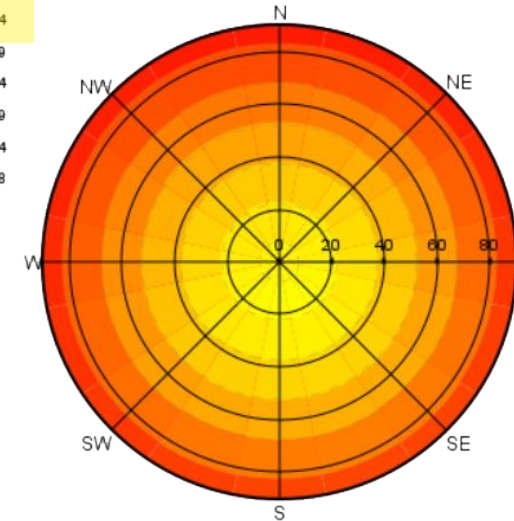


Future weather file

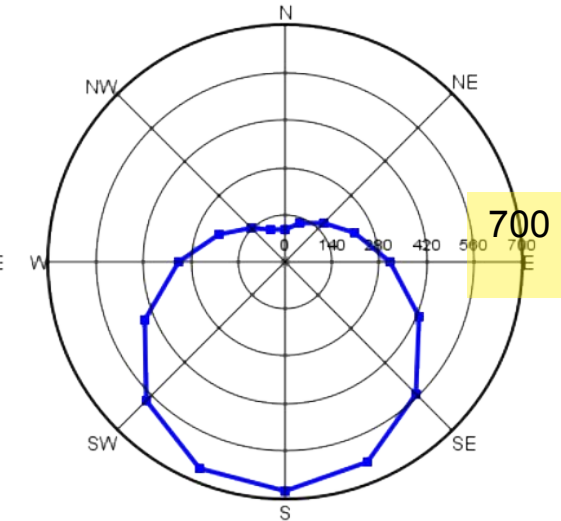
Mean Temperature [°C]:	8.3	Mean Relative Humidity [%]:	80.3
Max. Temperature [°C]:	28.6	Max. Relative Humidity [%]:	100
Min. Temperature [°C]:	-18.4	Min. Relative Humidity [%]:	27.3
Mean Wind Speed [m/s]:		4.3	
Normal Rain Sum [mm/a]:		982.8	

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Solar Radiation Sum [kWh/m²a]

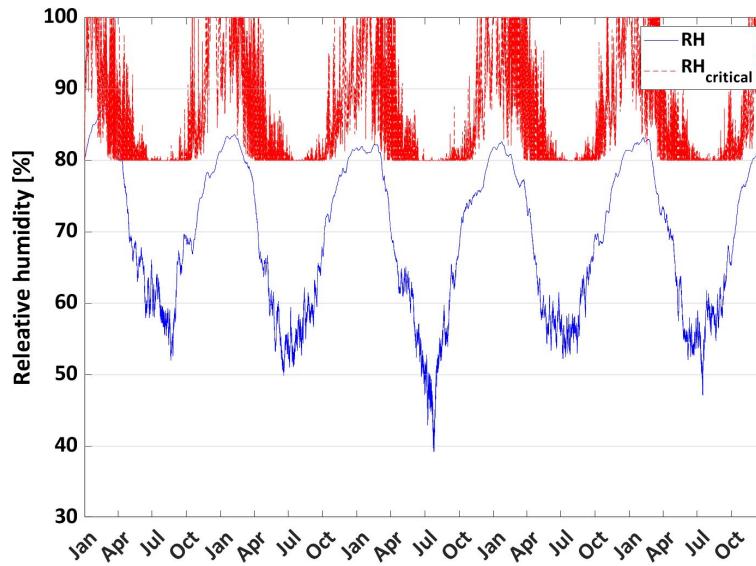


Driving Rain Sum [mm/a]

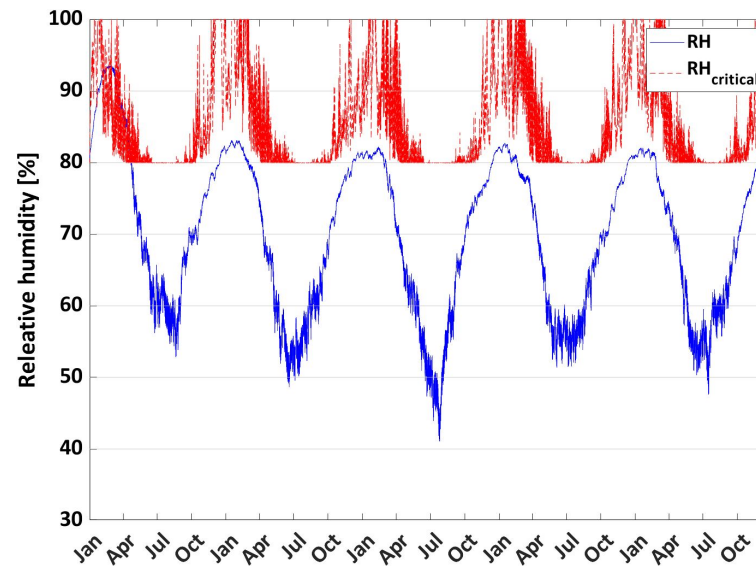


RESULTS IMPROVED VERSION

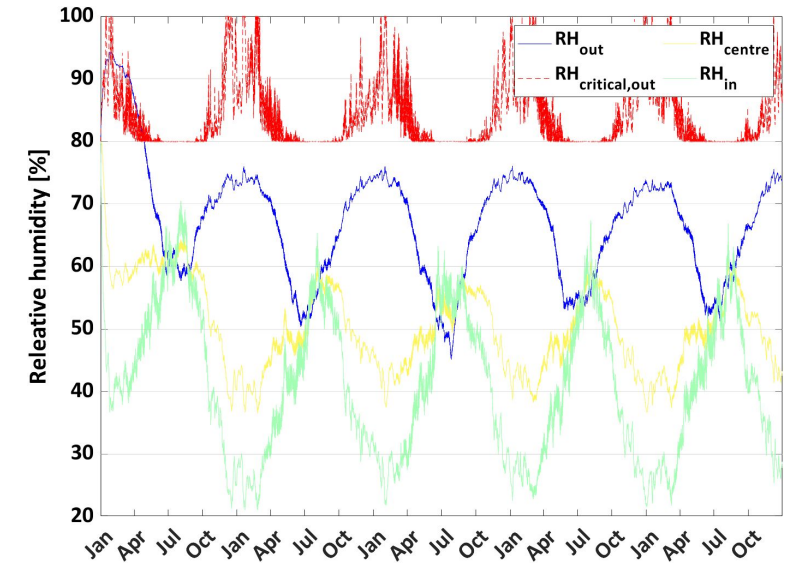
Layer 1 - Spruce deck



Layer 4 - Rigid Underlay board WF-HD

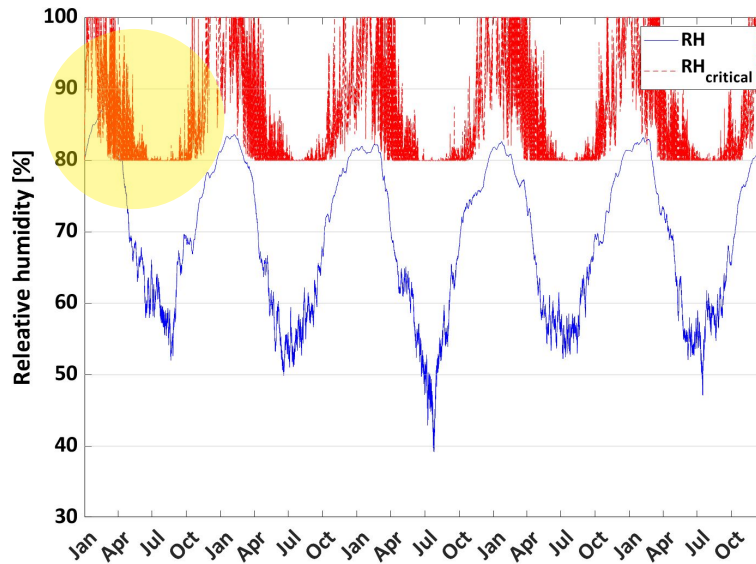


Layer 5 - Mineral wool insulation

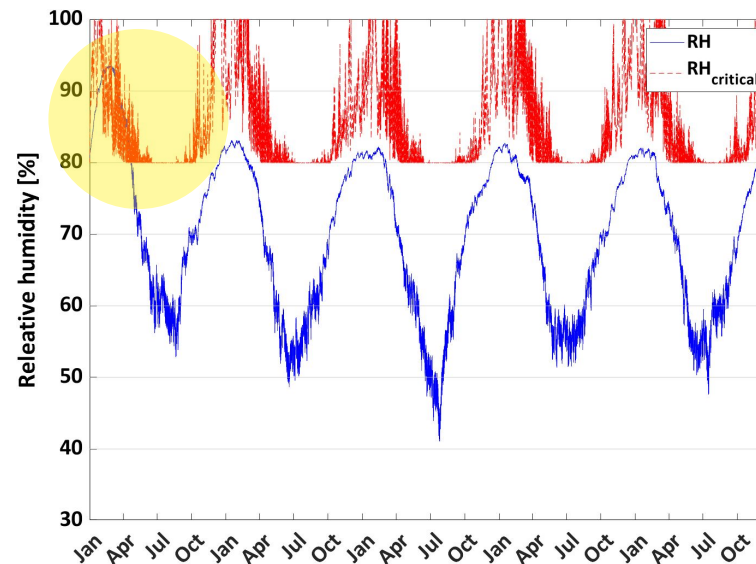


RESULTS IMPROVED VERSION

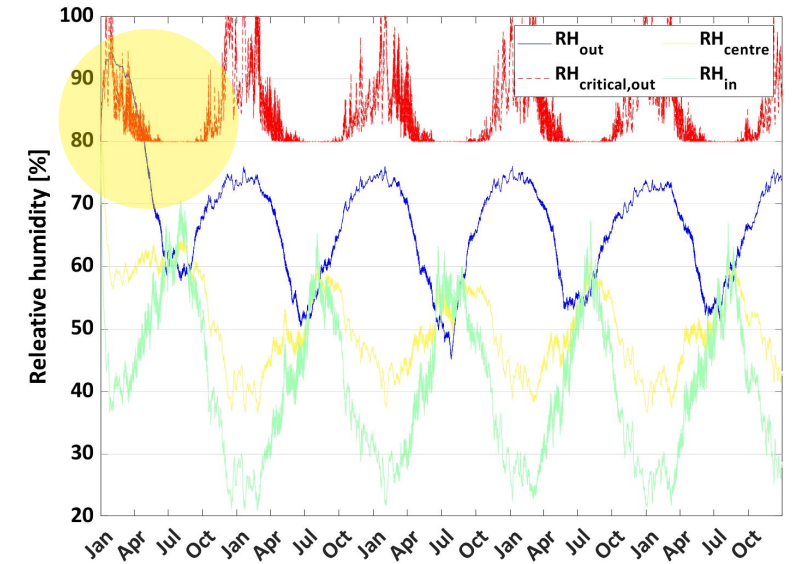
Layer 1 - Spruce deck



Layer 4 - Rigid Underlay board WF-HD

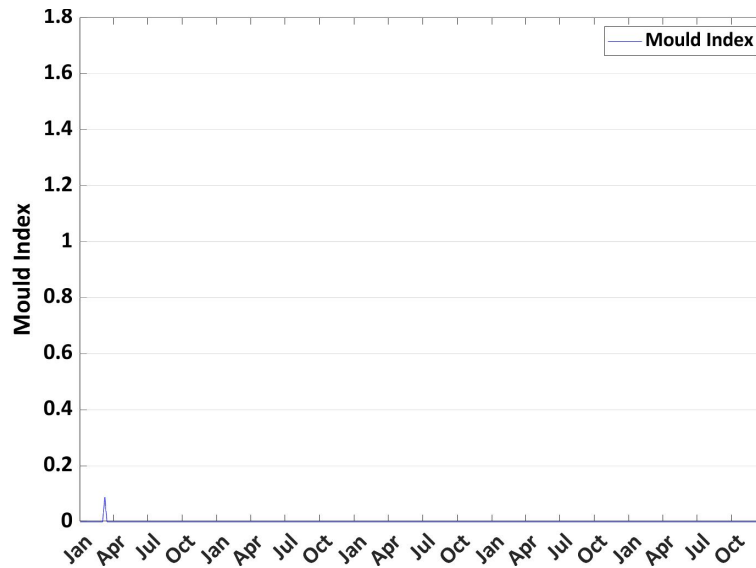


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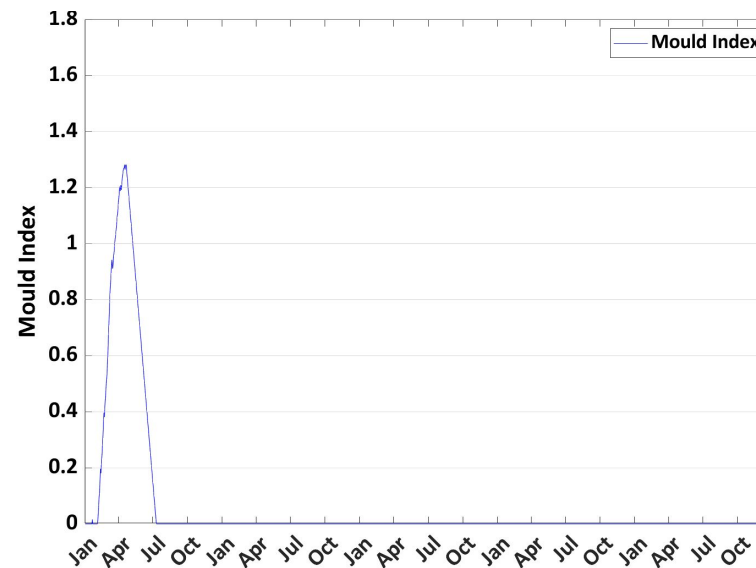


RESULTS IMPROVED VERSION

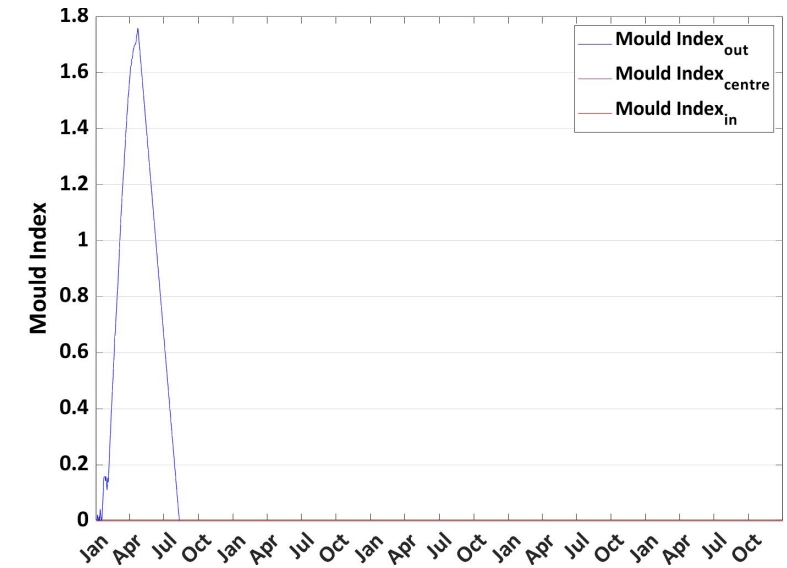
Layer 1 - Spruce deck



Layer 4 - Rigid Underlay board WF-HD



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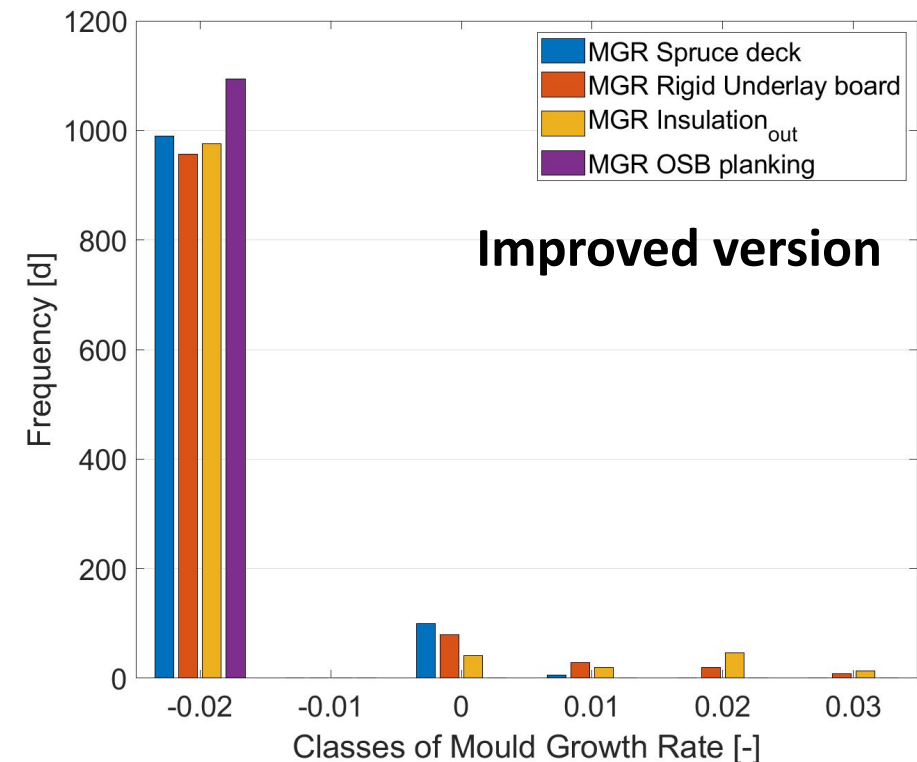
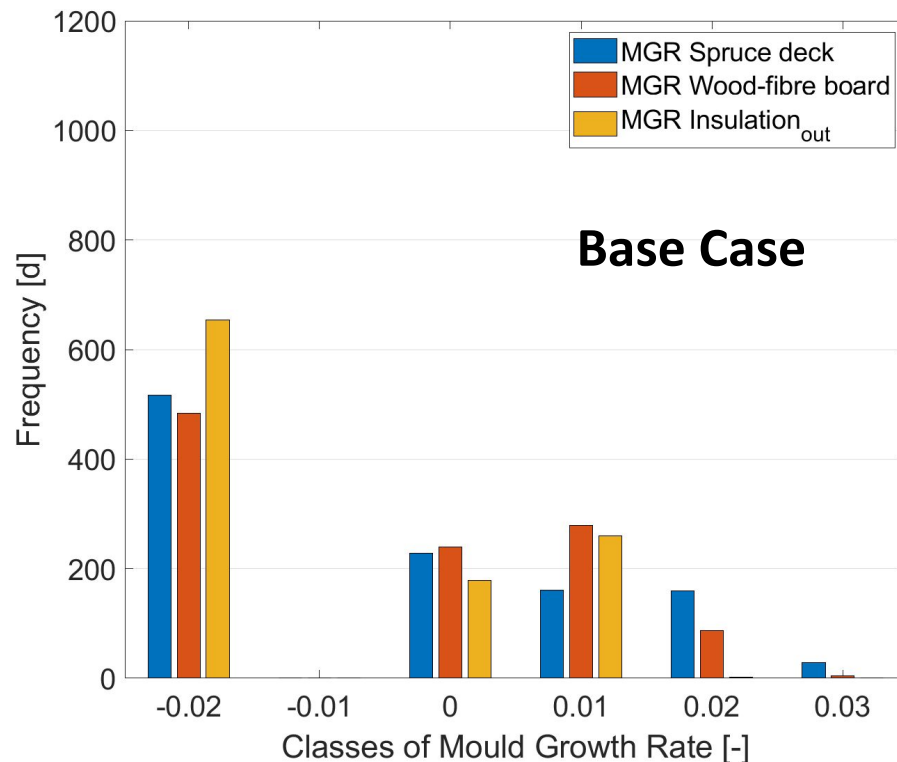
Tab 2. Summary of Mould Growth Assessment for the sensible layers. Improved version, five years, incl. initial conditions.

Layer	RH _{max} [%]	RH _{mean} [%]	T _{mean} [°C]	MI _{max}	MI _{mean}	MGR <0 [h-%]
Spruce deck	87.81	69.49	10.35	0.09	0.00	89.05
Rigid Underlay WF board	93.58	70.27	10.87	1.28	0.06	89.65
Mineral Wool (outside)	94.31	67.17	11.85	1.76	0.10	92.88
OSB 3	80.00	42.44	20.67	0.00	0.00	99.89

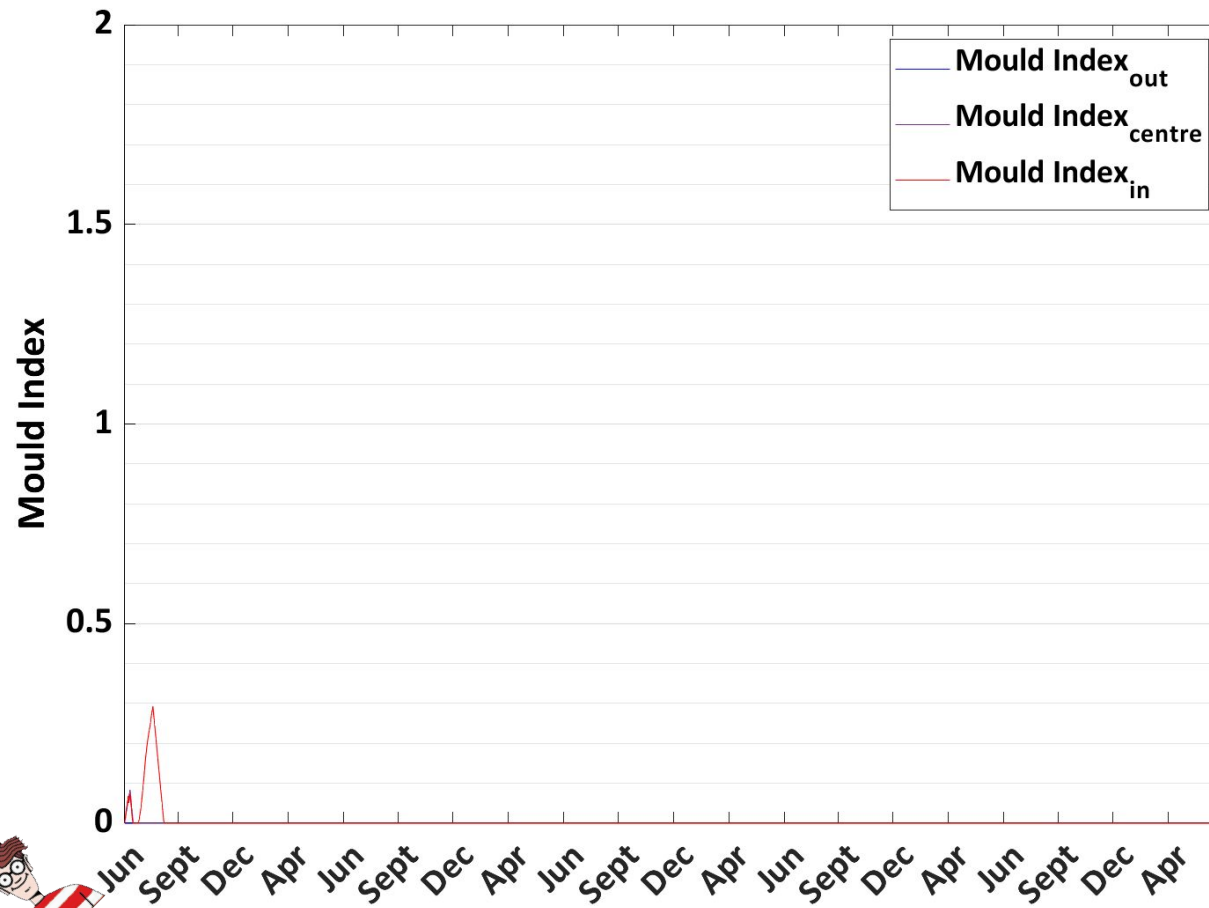
SUMMARY

- Max Mould Index Base Case
- Max Mould Index Improved V.

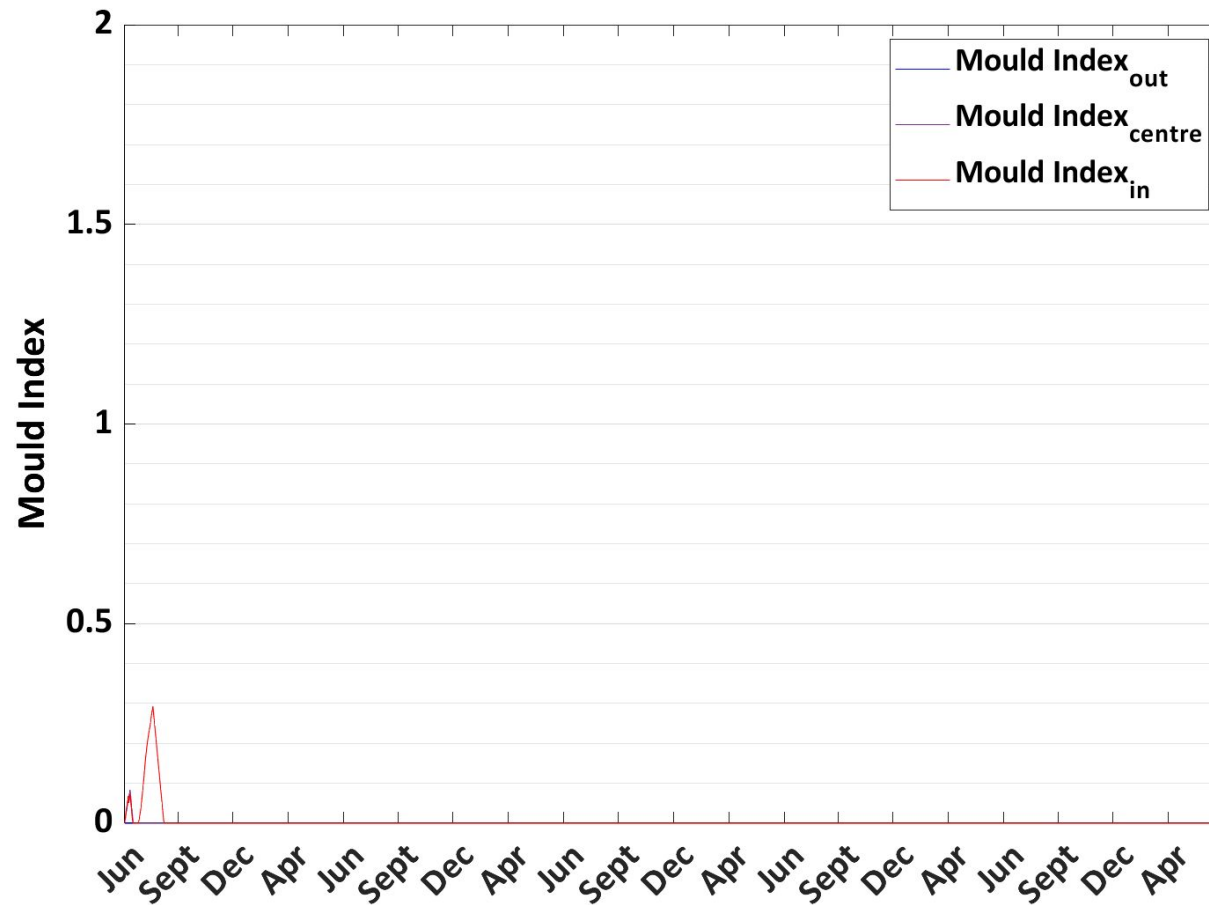
1.81 (WF board)
1.76 (MW outside)



KEY LESSON



KEY LESSONS



+ VENTILATION !!!

REFERENCES

- A. Hukka and H. Viitanen, 'A mathematical model of mould growth on wooden material', *Wood Sci. Technol.*, vol. 33, pp. 475–485, Dec. 1999, doi: 10.1007/s002260050131.
- L. Gullbrekken, T. Kvande, and B. Time, 'Ventilated wooden roofs: Influence of local weather conditions - measurements', *Energy Procedia*, vol. 132, pp. 777–782, Oct. 2017, doi: 10.1016/j.egypro.2017.10.029.